Primitive Recursion on Algebraic Datatypes

Lecture Notes Transcription

Course: Interactive Theorem Proving (7CCSACTP)

1. Context & Binary Tree Datatype Definition

Previously Covered:

- Isabelle interface & basic reasoning (FW, BW, `subst`, induction on nats)
- Lists (recursive functions, structural induction)
- Algebraic datatypes (`nat`, `list`, `tree`)

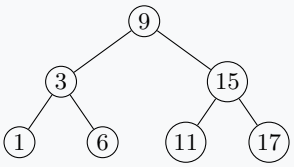
This Video:

- Primitive recursion on user-defined algebraic datatypes

Datatype Definition in Isabelle:

```
datatype 'a tree = Leaf | Node "'a tree" 'a "'a tree"
```

Example Binary Search Tree (BST):



2. Primitive Recursive Functions: `tree_set`

Set of Elements in a Tree:

$$\texttt{tree\_set Leaf} = \emptyset$$
$$\texttt{tree\_set (Node l a r)} = \text{insert } a\ (\texttt{tree\_set } l \cup \texttt{tree\_set } r)$$

Evaluation Trace on Example Tree:

$$\begin{aligned}\texttt{tree\_set } T &= \text{insert } 9\ (\texttt{tree\_set } T_3 \cup \texttt{tree\_set } T_{15}) \\ &= \text{insert } 9\ (\text{insert } 3\ (\texttt{tree\_set } T_1 \cup \texttt{tree\_set } T_6) \cup \text{insert } 15\ (\texttt{tree\_set } T_{11} \cup \texttt{tree\_set } T_{17})) \\ &= \{1,3,6,9,11,15,17\}\end{aligned}$$

3. Inorder Traversal of Trees

Definition of Inorder Traversal:

$$\texttt{inorder Leaf} = []$$
$$\texttt{inorder (Node l a r)} = (\texttt{inorder } l) \mathbin{@} [a] \mathbin{@} (\texttt{inorder } r)$$

BST Property: If the tree is a Binary Search Tree (where for every node, all left subtree elements are $<$ node and all right subtree elements are $>$ node), then `inorder` yields a **sorted list**:

$$[1,3,6,9,11,15,17]$$

Step-by-step Evaluation Trace:

$$\begin{aligned}\texttt{inorder } T_9 &= (\texttt{inorder } T_3) \mathbin{@} [9] \mathbin{@} (\texttt{inorder } T_{15}) \\ &= \big((\texttt{inorder } T_1) \mathbin{@} [3] \mathbin{@} (\texttt{inorder } T_6)\big) \mathbin{@} [9] \mathbin{@} (\texttt{inorder } T_{15}) \\ &= \big(([] \mathbin{@} [1] \mathbin{@} []) \mathbin{@} [3] \mathbin{@} ([] \mathbin{@} [6] \mathbin{@} [])\big) \mathbin{@} [9] \mathbin{@} \ldots \\ &= ([1] \mathbin{@} [3] \mathbin{@} [6]) \mathbin{@} [9] \mathbin{@} ([11] \mathbin{@} [15] \mathbin{@} [17]) \\ &= [1,3,6,9,11,15,17] \quad \checkmark\end{aligned}$$